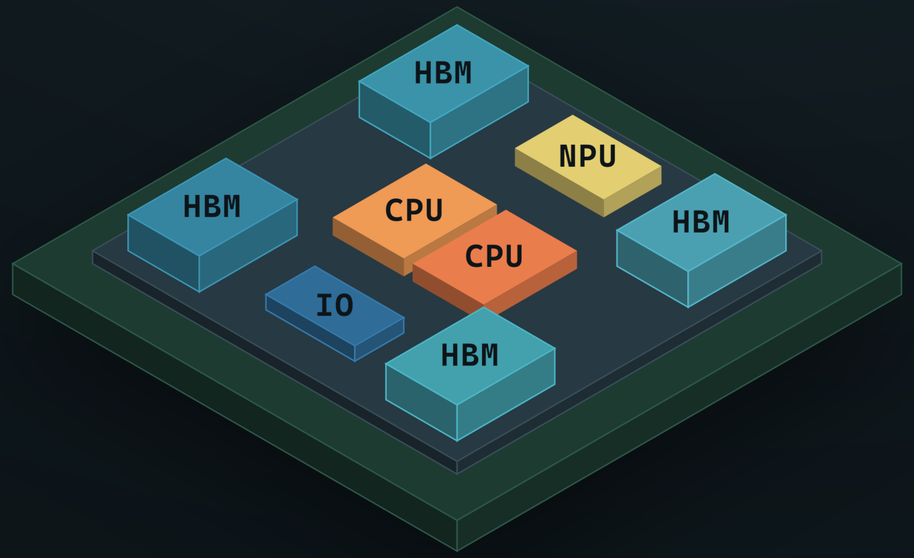
AI agents produce engineering findings faster than anyone can check them.

Lost	Findings scatter across chats, logs and branches.
Unchecked	Nobody can see what actually ran, or on which inputs.
Uncorrectable	When an input proves wrong, nothing shows what else it broke.



Our own chiplet visual, drawn for illustration: its colours are not measurements. First domain: chiplet layouts under thermal limits.

SENTILAB A lab designed so every result is signed, reproduced and correctable later.

Designed, demonstrated on a fixture, accepted in scoped slices.

DESIGNED

Labs propose — Several labs and a baseline submit candidate layouts.

The gate accepts — Only reproduced, verified results. No model decides.

The log records — Append-only. Each event hashed, chained and signed.

The Lab view shows — Only the verified prefix. It freezes when evidence breaks.

ACCEPTED IN SCOPED SLICES

PR53	Worker boot and source policy correction accepted	actual worker boot and source checks
PR51	Package view and lifecycle repairs reviewed and merged	source, merge and production HTTP checks
PR52	Scoped source slice has retained merge and CI evidence	retained merge and CI only
PR54	Admission and STOP corrections reviewed and merged	source, merge and CI only

DEMONSTRATED ON A TEST-KEY FIXTURE



Evidence breaks, the board freezes. A dropped event breaks the chain. The view stops at the last verified envelope; later envelopes are listed, never projected.

`Lab view · test-key interface fixture · illustrative values`

> Accepted slices are scoped: no blanket production acceptance, and no science or device grant follows from CI or a merge.

DATA Facts through 2026-10-31 15:05 UTC

engineering record evidence 2026-10-31 13:43 UTC, evidence 2026-10-31 13:55 UTC, evidence 2026-10-31 14:31 UTC, observed 2026-10-31 14:38 UTC

Withdraw a measurement. See what falls and what stands.

Candidate
A lab proposes a layout.

Reproduction
Re-run in a fresh process.

Acceptance
The gate accepts it.

Withdrawal
A measurement is withdrawn in a seeded-fault drill.

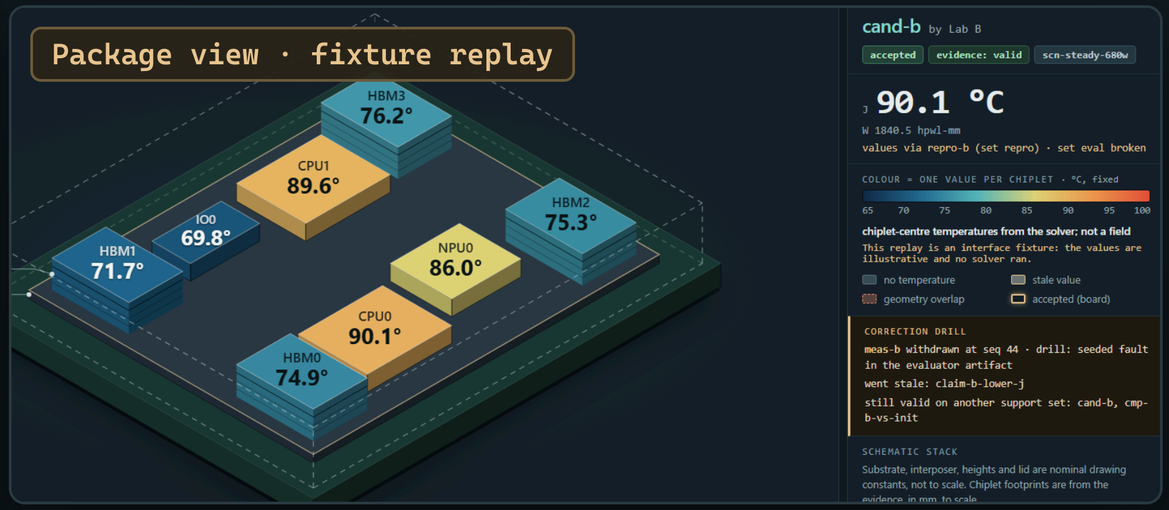
Propagation
Its dependents go stale. Independently supported results stand.

03 / EVIDENCE **What still stands**

9 valid · 1 stale · 1 withdrawn

WITHDRAWN	meas-b (seq 44: drill: seeded fault in the evaluator artifact)
WENT STALE	claim-b-lower-j ← meas-b
STILL VALID	cand-b via repro · cmp-b-vs-init via repro

NODE	KIND	VALIDITY	SUPPORT SETS	WHY / TIERS
meas-b	measurement	withdrawn	-	tier replayed
claim-b-lower-j	claim	stale	~~eval ✗~~	stale because meas-b
cand-b	candidate	valid	~~eval ✗~~ repro ✓	
cmp-b-vs-init	comparison	valid	~~eval ✗~~ repro ✓	
meas-init	measurement	valid	-	



Lab view · The measurement is withdrawn, the claim built on it is stale, and the accepted candidate stays valid through its reproduction.

Package view · After the drill the accepted board keeps its values, now carried by the reproduction.

Both views replay a test-key interface fixture: illustrative values, no solver ran. This drill shows the designed mechanism; the scientific correction is still open.

GB10: three separate attempts, each scoped exactly.

SLAB-018 attempt 1

completed · closed · historical

Original fixture preflight: train, save, reload, serve

20 steps | **40** HTTP responses | **112** checks

200 rows used once · **60** unique prompts

20/20 ADAPTER

0/20 BASE

Exact JSON on **six** source cases already seen in training, with labels supplied in the prompts. Not held-out.

494.2 s attributed: 314.2 s of stage time plus a 180 s prior reservation.

`historical charged attempt lifecycle including separate prior reservation`

first attempt record evidence 2026-10-31 07:53 UTC

SLAB-047

failed · closed

Serving attempt failed at startup, before inference

0 inference requests | **75** checks

A shared-object mapping failure was observed at startup. The original mount flags were not retained, so the suspected cause is unproved. The attempt is consumed.

79.6 s owned lifecycle, through the recorded cleanup.

`owned attempt lifecycle wall seconds through recorded cleanup`

failed attempt review evidence 2026-10-31 11:11 UTC

SLAB-047b

completed · closed

Serving fixture completed, with cleanup

20/20 EXACT JSON

not measured in this attempt BASE

Repeats **six** source cases already seen in training, with labels in the prompts. No base comparison here.

148.9 s whole owned lifecycle wall time, including cleanup. Not GPU time.

First client response **35.0 s**: client-observed latency, not throughput or time to first token; review had corrected a timeout too short for it. Launch to health 89.1 s. Native preflight 0.64 s, counted separately. 323 checks.

serving attempt review last response 2026-10-31 11:40 UTC, cleanup confirmed 2026-10-31 11:40 UTC, reviewed 2026-10-31 14:34 UTC

All three closed, no retry authorized. A single serving fixture completed; no learning or held-out generalization is shown.

The science is still open, with sharper questions.

WHERE THE OLD RESULT STANDS

Headroom remains inconclusive

The historical verdict, kept as recorded. No rerun result is claimed.

historical headroom record baseline 2026-10-31 07:50 UTC

Old evaluator temperatures and mesh observations: **historical observations · not corrected physics**.

claim corrections corrected 2026-10-31 14:34 UTC

Eight saved native fields analyzed; two representative cases independently checked. Hand-authored probes, not the headroom study. No new solve.

saved-field review reviewed 2026-10-31 14:45 UTC

REVIEWED HYPOTHESES, NOT RESULTS

REF-RECON accepted direction · execution gated

Reconstruct published references with orientation, transforms and uncertainty.

CAL-COUPLING accepted direction · execution gated

Check intended thermal equations and controlled paired diagnostics.

EVIDENCE-PROFILE source repairs required · no profiling grant

Measured phases, raw retention and forecast admission with runtime stops.

SCREEN-CONSTRAINED prospective protocol · not a result

Separate strict same-proxy, predeclared relaxed and Pareto tracks.

AGENT-PROPOSALS source slice reviewed · actual labs gated

Actual lab use requires provider/model/payer/entitlement admission and grant.

VIEW-LABELS dependent work · not a deliverable completion

Display matched reference/candidate scope and evidence limits.

NEXT · NATIVE GATE ZERO, DESIGN ONLY

8 primary calls + 4 fresh-process grid-64 replays

Design accepted 2026-10-31 15:02 UTC. Engine: TAP-pinned native HotSpot 6.0 under a new identity.

Executable source, input, host, window and grant binding: **pending**. Execution **not released**.

The envelope of up to 12 runs and 1,800 CPU-s is prospective, not a grant.

Prospective targets 75.48 / 115.94 °C at 45 mm: computational reference checks within a prospective 1 K tolerance. Physical accuracy not established.

Further published checks 75.47 / 94.89 °C: deferred to a later allocation, with no extra calls and no unseen holdout claimed. The old HotSpot 7 material-only 32-call plan: not requested.

gate-zero design accepted 2026-10-31 15:02 UTC

No calibrated temperature, silicon, optimality, MATCH/BEAT or universal-flatness claim.

None of the eight deliverables is fully accepted.

ORIGINAL SCOPE, NOT IMPLEMENTED COUNTS

41 tickets

234 requirements

104 CLI declarations

104 MCP declarations

Whole tickets complete: **0**. Deliverables fully accepted: **0** of **8**.

6 supplemental tracks add work outside the original tickets.

original scope baseline 2026-10-31 07:50 UTC

OPEN REPAIR QUEUE

Profiling PR55: four scoped closures; one open P2 (unknown method). Container CI failed in six cardinality subcases. 67 of 70 selected tests passed, three errored.

Parser PR56 parser needs one P2 repair. No native execution.

Bootstrap PR37 failed admission. A full conservative policy debit, not measured use, of 100 launches and 900 CPU-s is operator-reported · not independently verified.

Gate zero Design accepted; executable binding pending.

Study The original learning and sharing comparison, a real held-out worker evaluation and full CLI and MCP conformance remain open.

NEXT, PEOPLE AND COST

Priority: **research first**. Participant listener: one agent, the sole listener. Help room: announced · acceptance pending.

Current dollars: ***unknown***, not zero. USD 255.79 is a historical estimate, not current. Total CPU: ***unknown***.

Old scientific authority expired 2026-10-31 12:00 UTC. New execution **not released**.

Deck due 2026-11-01 18:00 UTC.

cost record estimated 2026-10-31 07:31 UTC

Scientific benefit **not established** | Full completion **not verified** | Public production **not fully accepted**

DATA Facts through 2026-10-31 15:05 UTC **PR55 corrective review** reviewed 2026-10-31 15:03 UTC **PR56 parser review** reviewed 2026-10-31 15:03 UTC **PR37 record** reviewed 2026-10-31 15:00 UTC **operator booking report** reported 2026-10-31 14:56 UTC **gate-zero design** accepted 2026-10-31 15:02 UTC **help-room record** reviewed 2026-10-31 14:59 UTC **scientific authority** expired 2026-10-31 12:00 UTC **delivery** due 2026-11-01 18:00 UTC

The evidence rules, as designed

SIGNED ENVELOPES

The core is the only writer. Each event becomes a canonical envelope: hashed, chained to the previous hash, and signed.

VERIFIED PREFIX

The view projects only the prefix that verifies. At the first gap, bad hash or conflict it stops; later envelopes are listed, never projected.

LIVE / STALE

LIVE needs verified progress, not just arrivals. A stall, a failed poll or a broken chain turns the label STALE and holds the board.

CORRECTIONS PROPAGATE

Withdrawing a measurement makes everything that depends on it stale. A result with another valid support set stays valid.

KNOWN LIMIT A server can hold back its newest envelopes while it keeps releasing older ones, and the view cannot tell. Only a later full replay shows the truth. Order is by sequence; timestamps are display only.

Engineering slices, scoped: no science or device grant.

PR53 `scoped slice accepted` evidence 2026-10-31 13:43 UTC

Worker boot and source policy correction accepted

head **58e3255** → merge **af1ac83**

Evidence: actual worker boot and source checks

The earlier actual-worker startup blocker closed on the corrected exact head. Retained actual boot, negative bad-export control, source/CSP checks and five green CI jobs support the scoped correction. Twenty-three changed blobs were bound to the merge.

- The earlier failed head stays historical.
- Nineteen passing records and two adjudicated redirect-expectation failures remain retained, not silently erased.
- No new full production-browser matrix or blanket production acceptance follows.

PR51 `scoped slice accepted` evidence 2026-10-31 13:55 UTC

Package view and lifecycle repairs reviewed and merged

head **e518546** → merge **1e6b97e**

Evidence: source, merge and production HTTP checks

Original three P2 issues closed in exact-head review; composition and five green CI jobs retained. Production HTTP checks bind both JavaScript assets and CSP to the merge. Both HTML files differ by the separately identified injected beacon markup.

- Production-browser observations remain separately operator-reported.
- HTTP/source evidence is not complete production custody, storage, cost or scientific acceptance.
- Reported J=118.4 C is a simulated scalar from an operator-identified run, not measured silicon or a newly certified thermal field.

PR52 `scoped slice accepted` evidence 2026-10-31 14:31 UTC

Scoped source slice has retained merge and CI evidence

head **eec0872** → merge **3e7c480**

Evidence: retained merge and CI only

- No fresh repository query was made while preparing this packet.

PR54 `scoped slice accepted` observed 2026-10-31 14:38 UTC

Admission and STOP corrections reviewed and merged

head **d63ccd7** → merge **1bdf584**

Evidence: source, merge and CI only

Both original admission/STOP P2 findings closed. The retained review records 23 supplied tests, 11 independent controls and 46 composition checks. Later merge binding verifies 17 changed blobs, identical head/merge trees and five successful CI jobs.

- No provider, oracle, model or device execution is released by this merge.
- A result completed before expiry but signed afterward was informational under the reviewed contract, not an extra blocking defect.
- The earlier pending-CI/merge observation remains in history; no live repository query was made for this packet.

> **SUPERSEDED HISTORY** 2026-10-31 14:31 UTC: scoped accepted · CI and merge pending. Replaced at 2026-10-31 14:38 UTC by: scoped slice accepted.

Open repairs, held findings and pending bindings.

PR55 · HEAD efe12c6

changes requested

P1	closed, scoped	captured result identity cross-check; foreign record and synthetic charge retained
P1	closed	four searches plus fresh reproduction, nine calls in admitted v3 fixture
P2	closed	terminal retained attempt state across supplied outcome paths
P2	closed	original 31 bytes retained through timeout and owned peer cleanup
P2	open	unsupported identifier accepted before one-search path; require refusal before calls

Tests: **67** of **70** selected passed; **three** errored for a missing native binary. Errors are not passes. Protocol tests **7/7**.

Existing CI: four jobs passed; the container job **failed** in **six** cardinality subcases, which need valid owned workspaces plus separate workspace-negative cases. Native execution: none.

> HISTORY, SUPERSEDED 2026-10-31 15:03 UTC The earlier review of head **18f0dca**, as recorded at 2026-10-31 14:31 UTC: “Two P1 and three P2 findings remain: an admitted long-SA reproduction split is not executed; measurements lack binding to their profiling attempt identity; a successful retained non-v4 attempt stays started; an unsupported method can PASS on the demonstrated one-search path; and 31 flushed worker stdout bytes without a newline disappear on timeout.”

PR55 corrective review reviewed 2026-10-31 15:03 UTC, published 2026-10-31 15:04 UTC

PR56 · HEAD 997269e

changes requested

PR56 parser needs one P2 repair: duplicate or malformed layers must fail before any summary.

20 tests passed, eight skipped, two retained-file controls; five existing CI jobs green. Native execution: none.

PR56 parser review reviewed 2026-10-31 15:03 UTC

PR37

source accepted · execution failed

Conservative policy debit, not measured use: **100** launches and **900 CPU-s**, operator-reported · not independently verified. The reported 0 launches and 0 CPU-s cover metered container work only. Runner CPU ***unknown***; money ***unknown***. No second attempt.

operator booking report reported 2026-10-31 14:56 UTC

GATE ZERO · STILL TO BIND

- TAP-referenced native6 commit and exact binary SHA256
- aarch64 and x86_64 build recipes and UVA notices
- raw and snapped reference geometry identities
- exact call manifest, numerical acceptance, allocation/host identity and current authority

STILL OPEN FROM THE ORIGINAL PLAN

The learning and sharing comparison, held-out worker evaluation, physical validation, model admission, the research harness, full CLI/MCP conformance.

Help room announced; participant completion remains unproved

PRIORITIES AND ROLES

`announced · acceptance pending`

Human direction through 2026-10-31 14:49 UTC: research first, a single participant listener, missing materials in parallel.

Priority
research first

Missing materials and requested UX
continue in parallel · acceptance pending

Participant listener and greeting
one agent

Duplicate approver
not requested

Human approval steps for ordinary help-room work
none

Automatic room-scoped admission
operator-reported, not independently verified

Agents keep their own identities; a human identity workaround is
not allowed

help-room record reviewed 2026-10-31 14:59 UTC

NOT YET SHOWN END TO END

The retained checkpoint of 2026-10-31 14:47 UTC observed a join, post and reply slice only.

- The observed join/post/reply slice is not a full fresh-account formal-claim/MCP/client/media flow or complete role audit.
- The exact complete skill bundle, every client runtime, media upload/release and automatic human-ticket flow are not completed by these reviews.
- A public app-beta HTTP response is not device enrollment or app acceptance.
- No participant roster, live online count or absence of joins is inferred from a bounded room read.
- Operator access promotion is not an independent fresh-account claim/post/connect/receive acceptance. Listener installation and missing-material delivery are not verified here.
- The five-file starter kit is not the complete requested three-skill bundle; documentation corrections remain requested.

RESEARCH SUPPLEMENT, DELIVERED PRIVATELY

Informed model critique: **static review complete**, delivered privately 2026-10-31 14:42 UTC. Its model was not executed or reproduced here.

`Cloud-provider compatibility assessment delivered`

published privately 2026-10-31 14:42 UTC

`assessment complete · deployment unproved`

- No archive transfer, hosted API connection, custom-weight admission, cloud runtime, provisioning or GB10 restart was tested.
- Regular storage lacks S3 Object Lock/Legal Hold; ETags are not guaranteed MD5. Versioning is not WORM retention.
- Managed custom weights are beta/on request; exact model/adapter eligibility, payer, price and entitlement remain unknown.
- The GB10 ARM64 image and compiled caches are not a proven portable x86 runtime; portable tensor/source reuse requires new platform checks.
- External GB10 integration as managed cloud-provider capacity is not documented. Private delivery is not scientific or deployment acceptance.

41 tickets, 8 deliverables: original scope, not implemented counts

234 requirements **104** CLI and **104** MCP declarations **0** whole tickets complete **0** deliverables fully accepted

D00	open · not accepted	005 006 028
D01	open · not accepted	005 006 028
D02	open · not accepted	002 006 007 028
D03	open · not accepted	015 028
D04	open · not accepted	007 010 028
D05	open · not accepted	020 021 028 032
D06	measured · not accepted	018 019 028 032 033
D07	open · not accepted	025 026 027 028
Shared	cross-cutting	001 003 004 008 009 011 012 013 014 016 017 022 023 024 029 030 031 034 035 036 037 038 039 040 041
Added	outside the original tickets	048 049 050 051 052 053

Ticket and deliverable statuses are as recorded at the baseline; added tracks are outside it. No whole ticket is complete and no deliverable is accepted.

`measured` Retained observations exist with the stated scope; success or scientific validity is not implied.

`open` Required acceptance work remains, or no completion evidence is retained.

`added` A supplemental track: added work, not part of the original ticket count.

COST, HONESTLY

Current total dollars

unknown

Invoice **not verified**. Comprehensive CPU ***unknown***.

Last retained estimate **USD 255.79**: historical, not a current balance.

cost record estimated 2026-10-31 07:31 UTC, cutoff 2026-10-31 15:05 UTC

The reported 0 launches and 0 CPU-s cover metered container work only, never runner CPU or money.

operator booking report reported 2026-10-31 14:56 UTC

Our own claims, corrected and dated.

First serving failure `cause unproved`

corrected 2026-10-31 14:34 UTC

A shared-object mapping failure was observed; original mount flags were not retained.

Later corrected configuration does not establish the old effective flags.

Serving time unit `metric corrected`

corrected 2026-10-31 14:34 UTC

148.9 whole owned lifecycle wall seconds, including cleanup.

No measured GPU busy/compute duration.

Budget fit `forecast only`

corrected 2026-10-31 14:34 UTC

A proportional forecast suggested a resource shortfall; means/reserve are not a hard bound.

No retrospective impossibility claim or current execution grant.

Participant onboarding `announced · not accepted end to end`

corrected 2026-10-31 14:34 UTC

Source/docs reviewed; exact guide, outside-account and actual-client flows remain unproved.

Keep transport source capability distinct from tested runtime access.

Old evaluator results `historical observations · not corrected physics`

corrected 2026-10-31 14:34 UTC

No replacement claim. Preserve exact source/results; calibration and new-evaluator error evidence are unresolved.

A correction replaces only its own claim. An earlier claim without a replacement is not restated here and is not reused.

Source findings behind the sharper questions.

Informed independent model critique preserved exact source evidence

reviewed 2026-10-31 14:36 UTC

Both frozen report/model hashes matched. Static comparison corroborates the convection-coupling lead in named upstream and vendored code. The one-cell centre selector differs from native 1/2/4-cell GRID_CENTER averaging; The model lacks a fail-closed convergence/nonfinite acceptance check before returning. Uncertainty combination, reference legality and fixed-G factorization assumptions need correction. The critique was subsequently delivered in the private research supplement.

- The independent model was not executed by this reviewer.
- Its numerical tables remain author-reported simplified-model outputs, not native HotSpot reproduction.
- Historical raw results, charges and original freeze remain intact; claim eligibility changes require explicit reviewed decisions.
- U=max is not a combined bound; increasing uncertainty cannot make the required MATCH tolerance easier. A failed reference does not prove the checker wrong.

Eight saved native fields analyzed; two representative cases independently checked

reviewed 2026-10-31 14:45 UTC

The parent analysis checked 112 saved-file hashes across eight grids and found same-index spreader-to-sink drops of 51.10–52.40 K. A separate parser checked 28 hashes for baseline16/32 only, corroborating 51.10 K and 51.60 K. Baseline32 has chip peak 118.89 C, spreader rise 70.11 K and sink rise 18.51 K at chip argmax index 527 (16,15).

- These are hand-authored 50 mm probes, not the historical headroom study or agent optimization. Only the two representative cases were independently re-parsed.
- Retained argv/raw dimensions are 16×16 and 32×32 with map=max, despite config defaults 64×64/centre. Native block-centre output is a different metric.
- Both retained configurations use r_convec 0.009; this rules out unscaled 0.1 for those configurations only. It does not establish the repair or physical fidelity.
- Raw output precision is 0.01 K. The same chip-layer argmax index is used across layers, not each layer maximum.
- Binary binding comes from the retained report and was not independently rehashed. No fresh solve, reproduction run or physical calibration occurred.

Reviewed hypotheses and source findings: no new solve, no reproduction run and no physical calibration.